RG09

Irrigation Projects and Canal Systems of Rajasthan

Irrigation projects matched to beneficiary districts have been asked in effectively every recent paper - 2013, 2016, 2018, 2021, 2023 and 2024. Budget 2-3 questions. IGNP alone can give one; traditional water harvesting gave two full questions in 2024.

This is the every-paper chapter of Rajasthan Geography. A dry state with one perennial river has had to import water from four directions, and every one of those schemes has a name, a river and a list of beneficiary districts that RPSC matches against each other. If you learn nothing else in Rajasthan Geography, learn the Indira Gandhi canal system and the project-river-district table.

Irrigation of Rajasthan

IGNP
- Kanwar Sen 1948, renamed 1984
- Harike Barrage, Sutlej-Beas
- Rajasthan Feeder 204 km + Main 445 km
- Stage-I 189 km, Stage-II 256 km to Mohangarh
- 7 lift canals, all on the left bank

Northern canals
- Gang Canal 1927 - oldest
- Bhakra-Nangal, Gobind Sagar
- Sidhmukh-Nohar (Ravi-Beas)
- Yamuna share, 1994 MoU

Chambal chain
- Gandhi Sagar (Mandsaur, MP)
- Rana Pratap Sagar (Rawatbhata)
- Jawahar Sagar (Kota)
- Kota Barrage - no power
- 50:50 Rajasthan-MP

Big dams
- Bisalpur - Banas, Tonk, drinking water
- Jawai - lifeline of Marwar, Pali
- Mahi Bajaj Sagar - longest, Banswara
- Parwan - Jhalawar, Baran, Kota
- Jakham - highest, Pratapgarh

Traditional harvesting
- Khadin, tanka, kund, nadi, toba
- Kui, paar, jhalara, baori
- Johad - Alwar, Rajendra Singh
- Chauka - Laporiya, Jaipur
- MJSA 2016, Jal Jeevan Mission 2019

New link
- ERCP - PKC-ERCP - Ram Jal Setu
- MoU 28 January 2024
- Renamed 22 January 2025
- 17 eastern districts

Why Rajasthan is an irrigation problem before it is an irrigation chapter

Understand the geography first and the projects will make sense. Rajasthan has about a tenth of India's area but a tiny share of its surface water. The Chambal is the state's only perennial river, and it flows through the south-east corner, far from the desert that needs it. Every other river is seasonal. So Rajasthan has solved its water problem by importing it: Sutlej and Beas water from Punjab in the north-west, Chambal water in the south-east, Mahi and Narmada water from Gujarat in the south, and now Chambal-basin water piped east under the Ram Jal Setu link.

- Wells and tube-wells irrigate the largest share of Rajasthan's net irrigated area - groundwater, not canals, is the state's main source.
- Canals come second and are concentrated in the Sri Ganganagar-Hanumangarh belt in the north-west.
- The Chambal is Rajasthan's only perennial river.
- The desert districts have no perennial river at all - which is exactly why the Indira Gandhi canal was built.
- Tanks and ponds matter in the south and south-east, where the land is uneven and rainfall is higher.

Which source dominates where

Source	Where it dominates	Why
Wells and tube-wells	Most of the state - largest share overall	Only reliable source where no canal reaches
Canals	Sri Ganganagar, Hanumangarh, Bikaner, Kota belt	IGNP, Gang Canal, Bhakra, Chambal canals

Source	Where it dominates	Why
Tanks and ponds	Southern and south-eastern districts	Broken relief holds run-off; higher rainfall

Indira Gandhi Nahar Pariyojana – the spine of the chapter

This is the biggest canal project of its kind in the world by command area, and the single most examinable object in Rajasthan Geography. The idea belongs to one man. Kanwar Sen, then Chief Engineer of Bikaner state, proposed on 29 October 1948 that Punjab's surplus water be carried into the Bikaner desert. It was first called the Rajasthan Canal Project. The foundation stone was laid on 31 March 1958 by Union Home Minister Govind Ballabh Pant, and in 1984 the project was renamed after Indira Gandhi.

- Conceived by Kanwar Sen, Chief Engineer of Bikaner, on 29 October 1948.
- Original name: Rajasthan Canal Project. Renamed Indira Gandhi Nahar Pariyojana in 1984.
- Foundation stone laid 31 March 1958 by Govind Ballabh Pant, Union Home Minister.
- Water source: Harike Barrage, at the Sutlej-Beas confluence in Ferozepur district, Punjab.
- Gravity branches leave the RIGHT bank - the Rawatsar branch is the single left-bank exception.
- All seven lift canals take off from the LEFT bank.

The lengths – learn these four numbers cold

Component	Length	Note
Rajasthan Feeder	204 km	About 170 km in Punjab-Haryana, about 35 km in Rajasthan

Component	Length	Note
Indira Gandhi Main Canal	445 km	Entirely within Rajasthan
Total main course	About 649 km	204 + 445
Stage-I main canal	189 km	First stretch, from the Rajasthan border
Stage-II main canal	256 km	Remaining stretch, ends at Mohangarh, Jaisalmer

The path of the water, source to tail

Sutlej + Beas → Harike Barrage, Ferozepur (Punjab) → Rajasthan Feeder 204 km → Indira Gandhi Main Canal, Stage-I 189 km → Stage-II 256 km → Mohangarh, Jaisalmer

Stage-I vs Stage-II – the district split that gets matched

Stage-I	Stage-II
Main canal 189 km	Main canal 256 km
Command: Sri Ganganagar, Hanumangarh, northern Bikaner	Command: Bikaner, Jodhpur, Jaisalmer, Barmer, Churu, Nagaur
Older, fully developed; this is where waterlogging appeared	Newer, drier, ends at Mohangarh in Jaisalmer
Nearer the Punjab border	Runs deep into the Thar

ASKED IN RAS

Asked in RAS Pre 2013 – from where does the Indira Gandhi Canal take off? Answer: the HARIKE BARRAGE, at the confluence of the Sutlej and the Beas in Ferozepur district of Punjab. Do not write Bhakra or Ferozepur headworks; Ferozepur (Hussainiwala) headworks belongs to the Gang Canal.

YAAD RAKHO

Four numbers, one sentence: 204 feeder, 445 main, 649 total, and the main splits 189 + 256. Notice 189 + 256 = 445, so if you remember any two of the three you can rebuild the third. And the tail is Mohangarh - MO for MOhangarh, the MOst distant point.

The seven lift canals - old name, new name, districts

The Indira Gandhi canal runs along a ridge, so land lying above the canal cannot be irrigated by gravity. For that land, water is pumped up. Those pumping channels are the lift canals - all seven leave the left bank. Every one was later renamed after a person, and RPSC's favourite format is a match between the old name and the new one, or between the lift canal and its districts. Learn the table in both directions.

IGNP lift canals - present name, former name, districts benefited

Present name	Former name	Districts benefited
Kanwar Sen Lift Canal (head at Gharsana)	Lunkaransar Lift	Sri Ganganagar, Bikaner
Pannalal Barupal Lift Canal	Gajner Lift	Bikaner, Nagaur
Veer Tejaji Lift Canal	Bangarsar Lift	Bikaner
Chaudhary Kumbha Ram Lift Canal	Gandheli-Sahwa Lift	Hanumangarh, Churu, Jhunjhunu, Bikaner
Dr. Karni Singh Lift Canal	Kolayat Lift	Bikaner, Jodhpur
Guru Jambheshwar Lift Canal	Phalodi Lift	Jodhpur, Jaisalmer, Bikaner
Jai Narayan Vyas Lift Canal	Pokaran Lift	Jaisalmer, Jodhpur

- The Kanwar Sen Lift Canal is named after IGNP's originator and has its head at Gharsana.

- The Chaudhary Kumbha Ram Lift Canal is the drinking-water lifeline of the Churu-Jhunjhunu belt - the driest districts of the Shekhawati.
- The Rajiv Gandhi Lift Canal is separate from the seven: it was built to carry IGNP water to Jodhpur city for drinking supply.
- Bikaner appears in five of the seven lift canals - when in doubt in a match, Bikaner is usually one of the districts.
- Gravity branches = right bank (Rawatsar branch excepted). Lift canals = left bank. That one line answers a statement question.

ASKED IN RAS

Asked in RAS Pre 2018 - the Chaudhary Kumbha Ram Lift Canal, earlier the Gandheli-Sahwa Lift Canal, mainly supplies drinking water to CHURU and JHUNJHUNU. This is the single most asked lift canal.

ASKED IN RAS

Asked as a match in RAS Pre 2021 - present name to former name of the lift canals: Kanwar Sen = Lunkaransar, Pannalal Barupal = Gajner, Dr. Karni Singh = Kolayat, Chaudhary Kumbha Ram = Gandheli-Sahwa. Memorise those four pairs at minimum; add Veer Tejaji = Bangarsar, Guru Jambheshwar = Phalodi, Jai Narayan Vyas = Pokaran for safety.

What IGNP achieved - and the criticism you must be able to state

The canal turned parts of the Thar green. Sri Ganganagar and Hanumangarh became wheat, mustard and cotton districts; dunes were fixed by afforestation along the canal; drinking water reached towns that had lived on tankas. But the same success created the problem. The desert soil is sandy above and impermeable below, and unlined canals plus flood irrigation pushed the water table up. The result is 'sem' - waterlogging and salinity - in the old Stage-I command. Every statement-based question on IGNP is built on this pair of facts.

- Benefits: irrigation across the north-western districts, drinking water to Bikaner, Churu, Jhunjhunu, Jodhpur and Barmer towns.
- Benefits: dune stabilisation and afforestation belts along the canal; new settlement in Ganganagar-Hanumangarh; industry and dairying followed.
- Criticism: waterlogging and salinity ('sem') in the Stage-I command of Sri Ganganagar and Hanumangarh - IGNP's best-known ecological cost.
- Criticism: heavy seepage from unlined stretches, flood irrigation instead of sprinkler, and rising water tables.
- Criticism: loss of traditional grazing tracts and change in the desert ecology; spread of water-borne disease in newly irrigated tracts.
- The command lies in the desert basin drained by no perennial river - never place IGNP in the Chambal basin.

- Wrong-statement drill 1: 'the entire IGNP command lies in the Chambal basin' - false; the Chambal is in the south-east and has its own project.
- Wrong-statement drill 2: 'the Kanwar Sen Lift Canal serves Banswara' - false; Banswara is in the Mahi basin, far outside the IGNP command.
- Rule of thumb: if any southern or south-eastern district is paired with an IGNP component, that pairing is wrong.

The northern canals - Gang Canal, Bhakra, Sidhmukh-Nohar, Yamuna

Before Indira Gandhi's canal there was Ganga Singh's. The Chhappaniya famine of 1899-1900 devastated Bikaner state, and Maharaja Ganga Singh negotiated Sutlej water from Punjab to protect his territory. The Gang Canal opened on 26

October 1927 and is Rajasthan's oldest canal irrigation project. Sri Ganganagar - the district named after him - owes its agricultural prosperity to it.

- Gang Canal: opened 26 October 1927, Rajasthan's oldest canal; built by Maharaja Ganga Singh of Bikaner after the Chhappaniya famine.
- It draws Sutlej water at the Ferozepur (Hussainiwala) headworks and irrigates Sri Ganganagar district.
- Total length about 129 km, of which only about 17 km lies inside Rajasthan - the rest is in Punjab.
- Bhakra-Nangal: on the Sutlej at Bhakra in Bilaspur district, Himachal Pradesh; its reservoir is Gobind Sagar.
- Bhakra is a joint Punjab-Haryana-Rajasthan project; Rajasthan's share is about 15.2%, used in Hanumangarh and Sri Ganganagar.
- Sidhmukh-Nohar project: uses Ravi-Beas water; serves Nohar and Bhadra in Hanumangarh and Sidhmukh and Taranagar in Churu; assisted by the European Union and later NABARD.
- Yamuna water: the 1994 MoU among Haryana, Uttar Pradesh, Rajasthan, Himachal Pradesh and Delhi fixes Rajasthan's share.
- The Gurgaon Canal carries Yamuna water to Bharatpur; a 2024 Haryana-Rajasthan MoU covers Yamuna water for Churu, Sikar and Jhunjhunu.

ASKED IN RAS

Asked in RAS Pre 2016 - the oldest canal irrigation project of Rajasthan is the GANG CANAL. If the paper gives a year, it is 1927; if it gives a ruler, it is Maharaja Ganga Singh; if it gives a district, it is Sri Ganganagar.

TRAP

Three headworks, three canals, and students mix them. HARIKE Barrage (Sutlej-Beas confluence) feeds the INDIRA GANDHI canal. FEROZEPUR / Hussainiwala headworks feeds the GANG CANAL. BHAKRA dam on the Sutlej in Himachal impounds GOBIND SAGAR and feeds the Bhakra system. All three are Sutlej-family, all three serve Sri Ganganagar-Hanumangarh, and all three are different structures.

The Chambal Valley Project - one river, four structures, in order

The Chambal is Rajasthan's only perennial river and it is shared with Madhya Pradesh on a strict fifty-fifty basis. Four structures were built on it, and the exam's favourite format is to give them jumbled and ask for the upstream-to-downstream order. Learn the sequence once and it never goes: Gandhi Sagar, Rana Pratap Sagar, Jawahar Sagar, Kota Barrage.

Chambal Valley Project - upstream to downstream

Gandhi Sagar (Mandsaur, MP) → Rana Pratap Sagar (Rawatbhata, Chittorgarh) → Jawahar Sagar (Kota) → Kota Barrage (Kota)

The four structures

Structure	Location	Point to remember
Gandhi Sagar	Mandsaur district, Madhya Pradesh	First and largest reservoir of the chain - lies in MP, not Rajasthan
Rana Pratap Sagar	Rawatbhata, Chittorgarh	Rajasthan Atomic Power Station stands at the same site
Jawahar Sagar	Kota	The pick-up dam between Rana Pratap Sagar and Kota Barrage

Structure	Location	Point to remember
Kota Barrage	Kota	No power house - it only diverts water into the canals

- Chambal Valley Project is a 50:50 joint venture of Rajasthan and Madhya Pradesh.
- Kota Barrage feeds the Right Main Canal and the Left Main Canal.
- Those canals irrigate Kota, Bundi, Baran, Sawai Madhopur and Tonk.
- Bisalpur is NOT part of this project - it is an independent scheme on the Banas in Tonk.

TRAP

Kota Barrage generates NO hydro power. The other three structures do. A statement question is built on exactly this. Second point: Gandhi Sagar is in Madhya Pradesh - if a question says all four dams of the Chambal project are in Rajasthan, it is false.

Project to river to districts - the table the exam matches from

This is the table that has been asked in one form or another in every recent paper. Three columns, and RPSC can hide any one of them and ask for it. Read it across, then cover the third column and test yourself. The trick is that most projects are named after their river or their site, so the middle column is often free.

Major irrigation projects - project, river or source, benefiting districts

Project	River / source	Benefiting districts
Indira Gandhi Nahar Pariyojana	Sutlej-Beas at Harike Barrage	Sri Ganganagar, Hanumangarh, Bikaner, Churu, Nagaur, Jodhpur, Jaisalmer, Barmer

Project	River / source	Benefiting districts
Gang Canal (1927, oldest)	Sutlej at Ferozepur headworks	Sri Ganganagar
Bhakra-Nangal	Sutlej (Gobind Sagar)	Hanumangarh, Sri Ganganagar
Sidhmukh-Nohar	Ravi-Beas	Hanumangarh, Churu
Chambal canals (Kota Barrage)	Chambal	Kota, Bundi, Baran, Sawai Madhopur, Tonk
Bisalpur	Banas (at Bisalpur, Todaraisingh tehsil, Tonk)	Jaipur, Ajmer, Tonk, Beawar, Kishangarh, Nasirabad - drinking water
Jawai ('lifeline of Marwar')	Jawai, a tributary of the Luni (at Sumerpur)	Pali, Jalore
Mahi Bajaj Sagar	Mahi (at Borkheda, Banswara)	Banswara, Dungarpur
Parwan Vrihad Pariyojana	Parwan, a tributary of the Kali Sindh (Jhalawar)	Jhalawar, Baran, Kota
Narmada Canal	Narmada, via Sardar Sarovar	Jalore, Barmer
Som-Kamla-Amba	Som	Dungarpur, Udaipur

- Bisalpur is the state's biggest drinking-water project, not primarily an irrigation dam.
- Jawai is the largest dam of western Rajasthan; begun by Maharaja Umaid Singh in 1946, completed 1957.
- Mahi Bajaj Sagar is a joint Rajasthan-Gujarat project and the longest dam of Rajasthan, about 3.1 km; the Kagdi pick-up weir downstream feeds its canals.
- Narmada Canal: Rajasthan is the tail-end beneficiary; 74 km of the canal lies in Rajasthan (532 km with the Gujarat stretch); it enters near Sanchore in Jalore.
- The Narmada command in Rajasthan is India's first canal command with 100% compulsory sprinkler and drip irrigation.

ASKED IN RAS

Two single-fact repeats. RAS Pre 2021 asked which river carries the Bisalpur dam - the BANAS, in Tonk district (not Sawai Madhopur). RAS Pre 2016 asked which district holds the Jawai dam, the 'lifeline of Marwar' - PALI.

ASKED IN RAS

Two more. RAS Pre 2023 asked which state joins Rajasthan in the Mahi Bajaj Sagar project - GUJARAT. RAS Pre 2024 asked which districts the Parwan Vrihad Pariyojana benefits - JHALAWAR, BARAN and KOTA. RAS Pre 2024 also ran the full project-to-district match: Jawai = Pali and Jalore, Parwan = Jhalawar-Baran-Kota, Narmada = Jalore and Barmer, Som-Kamla-Amba = Dungarpur and Udaipur.

The superlatives - dam to district

Other dams worth a line each

Dam	District / river	Why it is asked
Jakham	Pratapgarh, on the Jakham (a Som tributary)	Highest dam of Rajasthan, about 81 m
Panchana	Karauli	Biggest earthen dam; built at the confluence of five rivers
Mahi Bajaj Sagar	Banswara, on the Mahi	Longest dam of Rajasthan, about 3.1 km
Meja	Bhilwara, on the Kothari	Standard match-the-river item
Gudha	Bundi, on the Mej	Standard match-the-district item
Morel	Sawai Madhopur	Do not confuse with Bisalpur, also on the Banas system
Chhapi	Jhalawar	Hadoti group
Orai	Chittorgarh, on the Orai	Named after its own river
Sardar Samand	Pali	Marwar group with Jawai

Dam	District / river	Why it is asked
Bandh Baretha	Bharatpur, on the Kakund	Eastern Rajasthan
Isarda	Tonk-Sawai Madhopur boundary, on the Banas	Under construction; to supply Dausa and Sawai Madhopur

YAAD RAKHO

Three superlatives, three different dams - do not merge them. HIGHEST is Jakham (Pratapgarh). LONGEST is Mahi Bajaj Sagar (Banswara). BIGGEST EARTHEN is Panchana (Karauli). J-M-P: Jakham high, Mahi long, Panchana earthen.

Ram Jal Setu Link Project - the ERCP under its new name

Eastern Rajasthan has the people but not the water; the Chambal basin has the water but the terrain to lift it was expensive. The Eastern Rajasthan Canal Project was designed to move surplus monsoon flows of the Chambal and its tributaries into the deficit basins of the east. It has changed names twice, and the exam will test the names before it tests the engineering. Learn the chronology.

The naming chronology - this is the likely question

ERCP (Eastern Rajasthan Canal Project) → National project status, December 2022 → Tripartite MoU, 28 January 2024 - Centre, Rajasthan, Madhya Pradesh - becomes PKC-ERCP → Renamed Ram Jal Setu Link Project, 22 January 2025

- Full form of PKC: Parvati-Kali Sindh-Chambal link, integrated with ERCP under the National Perspective Plan for river interlinking.
- Donor rivers: the Chambal and its tributaries Parvati, Kali Sindh, Mej and Chakan.
- Recipient (deficit) basins: Banas, Banganga, Gambhiri and Parbati.
- It serves 17 eastern districts and about 40% of Rajasthan's population - drinking water plus irrigation.
- It was given national-project status in December 2022; the tripartite MoU with Madhya Pradesh and the Centre was signed on 28 January 2024.
- The Rajasthan government renamed it the Ram Jal Setu Link Project on 22 January 2025.

CURRENT LINK

Current link, as of September 2026: the project is under implementation as the Ram Jal Setu Link Project, serving 17 eastern districts. The foundation stone was laid by the Prime Minister on 17 December 2024 and work on the Chambal aqueduct between Kota and Bundi began in 2025. Because district boundaries in Rajasthan were themselves redrawn in 2023-24 and again reviewed in December 2024, the exact list of 17 districts can be phrased differently in different official releases - answer '17 eastern districts' and 'about 40% of the population' rather than reciting a district list.

Traditional water harvesting - two full questions came from here in 2024

Before any canal existed, Rajasthan survived on structures that catch rain where it falls. This is not folklore for the exam - RPSC asked it as a straight match and as a statement set in 2024, and it is the kind of topic that repeats. The logic is simple: in the desert you either store rain under a roof (tanka, kund), spread it on a field (khadin, chauka), pond it in a depression (nadi, toba), or tap what has already soaked into the sand (kui, paar, jhalara). Group them that way and the table writes itself.

Traditional water harvesting structures – region and how each works

Structure	Region	How it works
Khadin (dhora)	Jaisalmer – developed by Paliwal Brahmins, 15th century	Earthen bund across the lower slope of a rocky catchment; crop is grown on the residual moisture
Tanka	Bikaner, Barmer, Jaisalmer, Churu	Covered underground tank in a house courtyard storing rainwater for drinking
Kund / Kundi	Western desert districts	Circular covered underground well; its saucer-shaped catchment is called the agor
Nadi	Desert villages generally	Village pond in a natural depression collecting run-off for people and animals
Toba	Thar – Jaisalmer, Barmer	Natural depression with its own catchment; holds water longer than a nadi
Kui / Beri	Beside a paar or a tank	Narrow, deep pit that taps water seeping through the sand
Paar	Western Rajasthan	The percolation area itself, where run-off soaks into sand and is drawn out through kuis
Jhalara	Jodhpur, Udaipur	Rectangular stepped tank fed by seepage from an upstream tank; community bathing and religious use
Baori	Bundi (city of stepwells), Abhaneri in Dausa (Chand Baori)	Stepwell – steps descend to the water table
Johad	Alwar and the eastern hills	Earthen check dam across a slope; stores run-off and recharges groundwater
Bandha	Alwar-Bharatpur	Earthen embankment across a drainage line to hold run-off
Chauka	Laporiya village, Dudu, Jaipur	Square bunded plots hold a shallow sheet of water on pasture; recharges and grows grass

- Johads were revived by Rajendra Singh and the Tarun Bharat Sangh, which brought the Arvari river back to life in Alwar.
- Naada and Saza Kuva are structures of southern Rajasthan - a Naada is a small check bund, a Saza Kuva a shared open well.
- Mukhyamantri Jal Swavlamban Abhiyan was launched on 27 January 2016 at Nimbi Jodha in Nagaur.
- Jal Jeevan Mission - the central tap-water-to-every-rural-household mission - began on 15 August 2019.
- Zing is a Ladakh meltwater tank, not Rajasthani - it has been used as the odd one out.

ASKED IN RAS

Asked in RAS Pre 2024 - who developed the khadin system? The PALIWAL BRAHMINS of Jaisalmer, in the 15th century. The same paper ran a statement set on tanka, kund and jhalara, all three correct, and a match linking khadin, kui/beri, toba and johad to their descriptions. Traditional harvesting was worth more than one question in 2024 alone.

YAAD RAKHO

Sort them by what they do and you cannot be tricked. UNDER A ROOF: tanka, kund. ON A FIELD: khadin, chauka. IN A HOLLOW: nadi, toba. FROM THE SAND: kui, paar, jhalara. ACROSS A SLOPE: johad, bandha. STEPS DOWN: baori.

REVISION RECAP

1. Chambal is Rajasthan's only perennial river; wells and tube-wells irrigate the largest share of the net irrigated area, canals second.
2. IGNP: conceived by Kanwar Sen on 29 October 1948 as the Rajasthan Canal Project; renamed in 1984.
3. Foundation stone 31 March 1958 by Govind Ballabh Pant; water from Harike Barrage at the Sutlej-Beas confluence, Ferozepur, Punjab.

4. Lengths: Rajasthan Feeder 204 km + Indira Gandhi Main Canal 445 km = about 649 km; Stage-I 189 km, Stage-II 256 km to Mohangarh.
5. Stage-I commands Sri Ganganagar, Hanumangarh and northern Bikaner; Stage-II commands Bikaner, Jodhpur, Jaisalmer, Barmer, Churu, Nagaur.
6. Gravity branches on the right bank (Rawatsar the exception); all seven lift canals on the left bank.
7. Lift canals: Kanwar Sen = Lunkaransar (head Gharsana); Pannalal Barupal = Gajner; Veer Tejaji = Bangarsar.
8. Chaudhary Kumbha Ram = Gandheli-Sahwa, drinking water for Churu and Jhunjhunu; Dr. Karni Singh = Kolayat; Guru Jambheshwar = Phalodi; Jai Narayan Vyas = Pokaran.
9. Rajiv Gandhi Lift Canal carries IGNP water to Jodhpur city; waterlogging and salinity ('sem') in Stage-I is IGNP's main criticism.
10. Gang Canal: oldest, opened 26 October 1927, Maharaja Ganga Singh, Ferozepur headworks, about 129 km with only 17 km in Rajasthan, irrigates Sri Ganganagar.
11. Bhakra on the Sutlej in Bilaspur (HP), reservoir Gobind Sagar, Rajasthan's share about 15.2%; Sidhmukh-Nohar uses Ravi-Beas water for Hanumangarh and Churu.
12. Chambal Valley Project is 50:50 Rajasthan-MP; order Gandhi Sagar (MP) - Rana Pratap Sagar (Rawatbhata) - Jawahar Sagar - Kota Barrage, which alone has no power house.
13. Kota Barrage's Right and Left Main Canals irrigate Kota, Bundi, Baran, Sawai Madhopur and Tonk.
14. Bisalpur is on the Banas in Todaraisingh tehsil, Tonk - the state's biggest drinking-water project, supplying Jaipur, Ajmer and Tonk.
15. Jawai (Pali, on a Luni tributary) is the 'lifeline of Marwar' and western Rajasthan's largest dam; 1946-1957.
16. Mahi Bajaj Sagar is a Rajasthan-Gujarat joint project at Borkheda, Banswara - longest dam of the state, with the Kagdi pick-up weir.
17. Parwan serves Jhalawar, Baran and Kota; the Narmada Canal serves Jalore and Barmer and is India's first 100% sprinkler-drip command. Superlatives: Jakham (Pratapgarh) highest, Mahi Bajaj Sagar longest, Panchana (Karauli) biggest earthen dam.
18. ERCP - PKC-ERCP after the 28 January 2024 tripartite MoU - Ram Jal Setu Link Project from 22 January 2025; 17 eastern districts, about 40% of the population.
19. Khadin = Paliwal Brahmins of Jaisalmer; tanka and kund store drinking water; kui, paar and jhalara tap seepage; chauka is Laporiya, Dudu, Jaipur.
20. Johad = Alwar, revived by Rajendra Singh; MJSA launched 27 January 2016 at Nimbi Jodha, Nagaur; Jal Jeevan Mission from 15 August 2019.